Detection Is Not Enforcement: Evidence Repair and Risk-Gated Release in Tool-Using ML Pipelines

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***Abstract*—Agentic ML pipelines act on evidence from tools they cannot verify. Evaluating defenses against corrupted evidence is hard because the correct outcome of a realistic agent task is rarely known, so harm is scored by another model or a rubric. We construct a testbed in which the correct outcome is a theorem: classification tasks from arithmetic statistics (Galois groups of number fields from Dedekind zeta coefficients) with a computable Bayes-optimal ceiling, an exact per-input posterior, and a decidable predicate for which observations are possible at all, so corruption harm is measured against an exactly defined ceiling rather than a model's judgment. Using it, we separate three questions that are usually conflated: whether corrupted evidence is *detectable*, whether its effects can be *removed*, and whether a pipeline's final artifact should be *released*. We show (i) a deterministic verifier with stated guarantee classes detects a decidable corruption class with zero false positives, together with a restricted impossibility result for what no label-symmetric verifier can detect; (ii) in a sequential pilot with an identical detector and identical detections, a warning-only policy was associated with a large regression on a combined attack (median accuracy $0.90 \rightarrow 0.15$, complete separation of runs) while silent verbatim repair of detected responses was associated with near-clean performance; a prespecified, randomized, interleaved replication is registered and under way; (iii) a fresh-holdout release gate bounds released risk relative to a trusted fallback against *arbitrary* upstream corruption, including the provably undetectable class, with a trusted release runtime that survived adversarial source audits. Guarantees are independent of the number of agents and of agent disposition; they are fallback-relative, conditional on stated trust assumptions, and do not establish general semantic correctness.**

I. Introduction

When a pipeline of language-model agents reaches a conclusion after consulting tools, retrieved documents, or other agents, three different things can go wrong with the evidence it acted on. The evidence may be *malformed*: values no honest source could produce. It may be *falsified but plausible*: real-looking values attributed to the wrong things. Or the pipeline's own *checks* may be compromised, so that it cannot detect its errors. Defenses are usually evaluated by asking whether a model *notices* such corruption, and the notice rate is reported as the security metric.

We argue, and then measure, that this metric answers the wrong question. Detecting corruption is a property of a checker; whether the detection changes the pipeline's output is a property of the enforcement policy that acts on it; and whether the output is safe to use is a property of a release decision that need not depend on either. These are separate layers with separate guarantees, and a pipeline can have the first without the second, or the third without the first.

Measuring this requires knowing the right answer. We use a family of synthetic tasks built from a theorem plus one stated modeling choice: for a Galois number field of degree d, the Dedekind zeta coefficients at an unramified prime are determined by the splitting type (a theorem), and the Chebotarev density theorem fixes the asymptotic frequency of each type over primes; our generator additionally draws splitting types *independently* across primes from that distribution (the modeling choice). Under that fully specified law each task has a Bayes ceiling that is exactly defined and computable to Monte Carlo precision, a posterior defined on every positive-probability input, and a decidable set of *attainable* coefficient values. The number theory is the instrument: it makes correctness, support violation, and label inversion independently checkable without a model-as-judge, and gives an exactly defined ceiling against which harm is measured (the regrets we report are empirical accuracies against a Monte Carlo estimate of that ceiling; exact conditional regret is available per supported input but is not the quantity in our tables). We do not claim the tasks resemble deployed agent workloads, and we say explicitly what does and does not transfer (§IX).

a) Contributions:

1) **A controlled testbed for evidence corruption with independently checkable outcomes** (§III): tool-using classification tasks with exact support constraints and posterior risk under a stated synthetic law; corruption channels spanning malformed outputs, recomputable falsehoods, and consistent relabeling; and a restricted label-symmetry indistinguishability result (Prop. 2) for verifiers whose trusted observations cannot distinguish class assignments. Validation against genuine number fields (LMFDB) assesses mathematical fidelity, not deployment-distribution equivalence.
2) **A policy-level evaluation separating detection from mitigation** (§V): with an identical deterministic detector and identical detection outcomes, two enforcement policies—warn and silently repair—are associated in a sequential pilot with outcomes ranging from a large regression to near-clean performance. A prespecified, randomized, interleaved study of the primary contrast

is registered and in progress (§V-B); pilot evidence is reported as exploratory association, not policy causation.

3) **A layered enforcement implementation with distinct guarantee classes** (§IV): evidence checks, conditional exact-response restoration, and a fresh-holdout release gate bounding population excess risk relative to a trusted fallback (Thm. 1), with a trusted release runtime whose obligations were adversarially audited. We expose the residual accepted-corruption class with a defense-aware relabeling attack and report both candidate and released-system performance.

These are contributions about measurement, policy behavior, and the composition of assurances. We do not claim new concentration bounds, a defense without trusted resources, or provably trustworthy multi-agent collaboration.

II. Threat Model and Trusted Resources

a) System: One or more LLM agents, each with a fixed role prompt, consult tools that expose a dataset: per-class coefficient distributions, labeled examples, a list of structurally meaningful indices, and a validation tool that executes a candidate rule on a held split. Agents produce an executable classification rule $h : \mathbb{Q}^N \to \{0, \ldots, k-1\}$.

b) Adversary: Controls tool response channels: it may return any bytes in place of an honest response, on any subset of calls, with any per-call probability. Channels studied: `data` (reattributed distributions, values attainable), `impossible` (values the Euler-factor formula forbids), `verification` (inflated validation, out-of-grammar), `verification_stealth` (inflated validation, grammar-conforming), `injection` (appended operator-style instructions), `combined` (all, mutually consistent), and `data_adaptive` (consistent relabeling across every evidence channel, validation left honest; §VI). A separate `saboteur` condition replaces an agent's role prompt with a compromised one.

c) Trusted computing base: The mediator between tools and agents; the dataset snapshot and any recomputation performed on it; the certification sampler, labels, evaluator, and release mechanism; a fixed fallback predictor. Out of scope: sandbox escape, credential theft, resource exhaustion, and covert channels other than tool content. These assumptions are substantive. Our contribution is to separate the guarantees they support from guarantees that would depend on agent behavior, not to remove them.

III. Testbed

For a Galois extension $K/\mathbb{Q}$ of degree d and an unramified prime p with residue degree $f \mid d$ and $g = d/f$ primes above it,

$$a_{p^k}(K) = \begin{cases} \binom{g+k/f-1}{g-1} & f \mid k, \\ 0 & \text{otherwise}, \end{cases} \tag{1}$$

and a_n is multiplicative. By Chebotarev, f is distributed as the order of a uniformly random group element. So the Galois group leaves provable statistical fingerprints in the coefficient vector, and the optimal decision rule is computable. Following Lee and Lee [10], we generate synthetic "virtual fields" from this law for nine two- or three-class tasks (degree 4 through 25); ceilings span 0.84–0.9998 at N=1000. Security experiments use sextic (C_6 vs. S_3, ceiling 0.9518 at N=1000; 0.8991 at N=200) and degree-21 (C_{21} vs. C_7:C_3, 0.9352), which leave headroom for harm to register.

a) The law, completely: Classes are uniform over the task's group list. For each prime $p \leq N$ (N=1000 in agent experiments, 200 in offline audits), a residue degree f is drawn independently from the class's Chebotarev measure (the distribution of element orders); the visible signature $(a_{p^k})_{p^k \leq N}$ follows from (1); composite a_n by multiplicativity. The class-conditional likelihood of a row is the product over primes of the pushforward mass of its signature (colliding signatures summed), and the posterior is Bayes' rule with the uniform prior on rows of positive mass; rows of zero mass have no posterior and are flagged inadmissible. The Bayes ceiling is exactly defined by this law and estimated by Monte Carlo (2×10^5 Bernoulli samples of the Bayes decision, binomial standard error $\approx 5 \times 10^{-4}$ at a ceiling of 0.95); per-feature mutual information is computed in closed form from the same pushforwards. The proven quartic criterion classifies 200 real LMFDB quartic fields (100 C_4, 100 C_2^2, coefficients from true decomposition data) at accuracy 1.000; this validates the coefficient arithmetic, not the independence assumption or the unramified restriction, which are modeling choices. The held-out families (degrees 14, 21, 25) appear in no publication we know of; this makes memorization of the *task instance* unlikely, not memorization of the underlying group theory.

IV. Three Layers, Three Guarantee Classes

A. Evidence layer: deterministic verification

A verifier interposed between tools and agents, importing nothing from agent or attack code (standard library plus one LP call, Prop. 1). Two scopes must be kept apart: the *label-agnostic subverifier* (theorem and schema checks) knows nothing that distinguishes candidate classes; the *complete mediator* additionally holds the trusted dataset snapshot and can recompute the validation tool, which is class-anchored information. Each check carries a guarantee class:

- **theorem**: (1) fixes the attainable values of a_{p^k} for *any* Galois field of degree d at unramified primes, and multiplicativity gives the per-coordinate attainable set of every composite a_n; a same-prime check requires one residue degree to explain all visible powers of p in a row; a cross-coordinate check enforces $a_{mn} = a_m a_n$ on example rows where every involved value renders losslessly. Values are checked as membership in the renderer's `%g`-image of the attainable set. Sound for honest generator output; complete against reported values outside the set. These are sound necessary checks, not a full row-admissibility decision (the offline audit module has one).
- **schema**: honest tools emit a fixed grammar; anything outside it (appended instructions, label/class-name incon-

sistency, forged refusals of valid requests) is flagged. Silent against grammar-conforming forgeries.
- **recomputation**: validation is deterministic under the harness's controlled execution, so the mediator re-derives it; any discrepancy is a proof of channel corruption. This is the only check that catches the format-mimicking `verification_stealth` attack.

The theorem class also includes a *joint* check on marginal reports, which closes a gap the coordinatewise checks leave open:

Proposition 1 (Same-prime closed-band realizability)**.** *Fix a prime p and a class. For each visible power k, the honest tool reports the class marginal of a_{p^k}, which is the image of one distribution q over residue degrees $f \mid d$ under $f \mapsto R(b_{d,k}(f))$, where $b_{d,k}$ is* (1) *and R the renderer. Call reported marginals $\{c_k\}$* closed-band realizable *if some latent mixture q has every rendered-value mass within τ of the report. This holds iff the linear program $q \geq 0,\ \sum_f q_f = 1,\ |\sum_{f:R(b_{d,k}(f))=v} q_f - c_k(v)| \leq \tau$ for all (k, v) is feasible (rendering collisions are summed in the constraint rows).*

"Realizable" here means realizable by a *population mixture* whose rendered report lies within the closed rounding tolerance (exactly $\pm\frac{1}{2000}$ for `%.3f`); the finite-sample version (integrality of nq_f for a class of n rows) is stronger and not implemented. The LP has as many variables as divisors of d. It is solved in floating point, but the verdict rests only on certificates verified in exact rational arithmetic: a feasible point is rationalized and every constraint re-checked, and a report is flagged only on an exactly verified Farkas certificate of infeasibility (obtained from a second LP); if neither verifies, or the solver fails, the report is accepted. It is a sound infeasibility detector with unresolved cases passing, not an exact decision procedure. Duplicate rendered keys (distinct raw values sharing a `%g` image) are aggregated with the band widened accordingly. The check is stateful across calls and transactional: a report is committed to the store only after it passes, and under repair the committed report is the honest replacement, so a rejected forgery cannot cause a later honest report to be flagged. What a flag establishes is that the accumulated evidence set is inconsistent with one sample—not, by itself, that the current response is the corrupt one. It is class-agnostic—q is free—so a consistent relabeling passes, as Prop. 2 requires, while individually attainable marginals that no mixture produces (e.g. $\Pr[a_p{=}d] = \frac{1}{2}$ with $\Pr[a_{p^2}{=}\binom{d+1}{2}] = 0$) are refused.

Detection of the implemented fixtures is computed, not sampled (Table I), identical on the three tasks tested, and scoped to those fixtures: it is not completeness for everything a channel could carry. Soundness ("honest output is never flagged") is proved for the arithmetic checks (Lean, below) and tested against the renderer for the grammar checks. The label-agnostic subverifier is kept free of class-distinguishing knowledge because a subverifier with it would be an oracle for the task and would void the measurement.

TABLE I
DETERMINISTIC DETECTION OF THE IMPLEMENTED ATTACK FIXTURES (COMPUTED; IDENTICAL ON ALL TASKS). HONEST OUTPUT IS NEVER FLAGGED.

channel	detected	guarantee
none (control)	no	— (soundness)
data / data_adaptive	**no, by design**	— (Prop. 2)
verification	yes	schema
verification_stealth	yes	recomputation
injection	yes	schema
combined	yes	schema
impossible	yes	theorem

Proposition 2 (Restricted label-permutation indistinguishability)**.** *Let a verifier observe class-indexed evidence transcripts, with trusted knowledge invariant under a permutation π of class labels, and let the attack consistently apply π to every class attribution in every observed transcript, changing nothing else. If the verifier has false-positive probability at most α in every admissible honest world, its detection probability against the permuted transcript is at most α.*

Proof. The attacked transcript has exactly the distribution of an honest transcript in the world πw, which is admissible by label-invariance of the trusted knowledge; apply the false-positive bound in that world. □

The proposition is a boundary, not a defense. It does *not* apply when the verifier has trusted labeled examples, exact labeled recomputation, or observes partial relabelings inconsistent across channels. Its detection matrix row is pinned by a regression test that fails if the verifier ever acquires class-distinguishing knowledge.

B. Protocol layer: enforcement policies

Given a detection, the mediator can *flag* (return the response verbatim with a diagnostic appended), *flag with routing* (diagnostic names the failed source and the channels that remain trustworthy), *block* (return a fixed code), or *repair* (return the trusted recomputation verbatim, with no notice). Block gives behavior-independent *containment* of the corrupt bytes; only repair admits a behavior-independent *restoration* guarantee:

Proposition 3 (Exact mediated-execution coupling)**.** *Consider a channel-attacked and a channel-clean execution with the* same *agent programs, initial prompts, orchestration policy, and exogenous inputs (so any compromised agents are present in both), with randomness modeled by coupled tapes. Let $\mathcal{R}$ relate their trusted environment states, initially satisfied, such that whenever agent states and visible histories agree and environment states are $\mathcal{R}$-related: (1) the next observable agent action is the same; (2) the attacked execution's mediated response to it equals the clean execution's response; (3) environment states remain $\mathcal{R}$-related afterwards; (4) termination and artifact extraction agree. If no other attacker-controlled observation or state transition reaches the agents, then every finite visible transcript prefix is identical, termination occurs at the same step, and final artifacts are identical.*

Proof. Induction on steps; (1)–(3) preserve the invariant, (4) closes. □

Detect-and-repair establishes (2) provided recomputation uses the correct snapshot and request and has no divergent side effects; for stateless deterministic tools on a fixed snapshot, $\mathcal{R}$ is equality. Two limits are part of the statement: the reference is the *channel-clean* execution with the same agents, so repair removes covered channel corruption even for badly behaved agents but does not turn a compromised agent into an honest one; and the conclusion is equality to that reference, not correctness or low risk.

The proposition is conditional on complete detection and exact restoration; tests verify byte equality of attacked-repaired and honest responses on every implemented detected fixture, which establishes hypothesis (2) for those fixtures and not for arbitrary bytes. The `data` channels are undetected and therefore unrepaired; that residual is what the third layer covers.

a) The three guarantees, kept separate: The layers compose into three statements with different characters, which we deliberately do not merge into one bound. (i) *Deterministic restoration* (Prop. 3): for the covered channel class, the attacked execution equals the channel-clean reference with the same agents; clean-relative, not a risk statement. (ii) *Statistical release safety* (Thm. 1): conditional on all pre-certification randomness, for arbitrary upstream corruption including all N agents compromised and attacks outside the covered class, $\Pr[L_D(h_{\text{out}}) > L_D(h_0) + \epsilon] \le \delta$; fallback-relative, per certification, with K the only proposal-count dependence, and no bound on $\Pr[\text{unsafe} \mid \text{a candidate was released}]$. (iii) *Label-symmetry boundary* (Prop. 2): for the label-agnostic subverifier only; a limit on detection, not on harm, and not on the gate's ability to refuse the resulting proposal.

b) What is machine-checked: Four statements are formalized in Lean 4 without Mathlib (`lean/`, 21 theorems, no `sorry`), each mapped to a claim: (i) the attainable set of (1) over divisors of d, with the theorem that 0 is attainable iff some divisor of d does not divide k (the arithmetic support lemma *conditional on* the Euler formula—the number theory itself is not formalized; this is the fact our first implementation violated); (ii) Prop. 2 in its zero-false-positive form, as an abstract statement about a verifier, honest set, and relabeling action; (iii) Prop. 3 as a trace induction over a deterministic transition system; (iv) the deterministic clause of Theorem 1: on the concentration event, any valid release has risk $\le L_D(h_0)+\epsilon$, including that $\epsilon \ge 0$ is necessary and that fail-closed is always safe. The concentration inequality is a hypothesis in (iv). None of these connects to the parser or runtime code; they fix the mathematics the code is tested against. The `data` channels are undetected and therefore unrepaired; that residual is what the third layer covers.

C. Output layer: fallback-relative release

The agent's artifact is executable. Its risk on fresh data is checkable regardless of how the agent reasoned or how corrupt its evidence was.

Theorem 1 (Fresh-holdout gate)**.** *Let $\mathcal{F}$ contain all pre-certification randomness. Let $h_1, \dots, h_K$ ($K \ge 1$) be candidate predictors and h_0 a fallback, all $\mathcal{F}$-measurable, pointwise, with identical certification and deployment semantics, and let $H = ((X_i, Y_i))_{i=1}^m$ satisfy $H \mid \mathcal{F} \sim D^m$, $m \ge 2$, with 0–1 loss ℓ and $0 < \delta < 1$. Let $Z_{ji} = \ell(h_j; X_i, Y_i) - \ell(h_0; X_i, Y_i) \in \{-1, 0, 1\}$, $\hat{\Delta}_j = \frac{1}{m}\sum_i Z_{ji}$, V_j the unbiased sample variance of $(Z_{ji})_i$, and*

$$r^H = \sqrt{2\log(2K/\delta)/m}, \tag{2}$$

$$r_j^B = \sqrt{2V_j \log(4K/\delta)/m} + \tfrac{14\log(4K/\delta)}{3(m-1)}, \tag{3}$$

$r_j = \min(r^H, r_j^B)$. A release rule may depend arbitrarily on H but releases only h_0 or a candidate with $\hat{\Delta}_j + r_j \le \epsilon$, $0 \le \epsilon \le 1$. Then, conditional on $\mathcal{F}$ and hence unconditionally, $\Pr[L_D(h_{\text{out}}) > L_D(h_0) + \epsilon] \le \delta$, and $\mathbb{E}[(L_D(h_{\text{out}}) - L_D(h_0))_+] \le \epsilon + \delta(1-\epsilon)$.

Proof. Work conditionally on $\mathcal{F}$. Spend $\delta/2$ on each inequality. One-sided Hoeffding on $[-1,1]$ with a union bound over K gives $L_D(h_j) - L_D(h_0) \le \hat{\Delta}_j + r^H$ for all j w.p. $\ge 1 - \delta/2$; the empirical-Bernstein inequality of Maurer and Pontil [12] (range 2) gives the same with r_j^B w.p. $\ge 1 - \delta/2$. On the intersection E, of probability $\ge 1 - \delta$, every accepted j has $L_D(h_j) - L_D(h_0) \le \hat{\Delta}_j + r_j \le \epsilon$, and rejection releases h_0 with excess $0 \le \epsilon$; this is the first claim. For the second, the positive excess is at most ϵ on E and at most 1 on E^c, and $\Pr(E^c) \le \delta$, so its expectation is at most $\epsilon + \delta(1-\epsilon)$. Average over $\mathcal{F}$. □

Repeated certifications each carry their own δ; a campaign of T releases has failure probability at most $\sum_t \delta_t$, and re-certifying until something passes is not covered.

The variance term is what makes the gate usable against a strong fallback. Paired 0–1 differences are mostly zero when the two predictors are close, so $V_j \ll 1$ and $r^B \ll r^H$. A worked instance, with the paired counts that determine everything: a genuine agent rule (sextic, clean pilot run) against the theorem fallback on one fresh m=2000 holdout gives $(n_+, n_0, n_-) = (72, 1906, 22)$, i.e. disagreement on 4.7% of rows, $\hat{\Delta} = 0.025$, $V = 0.046$ (holdout accuracies 0.9345 vs. 0.9595). With K=1, δ=0.05: $r^H = \sqrt{2\ln 40/2000} = 0.0607$ and $r^B = \sqrt{2 \cdot 0.046 \ln 80/2000} + 14\ln 80/(3 \cdot 1999) = 0.0245$, a $2.5\times$ reduction, and $\hat{\Delta} + r^B = 0.0495 \le \epsilon = 0.05$: the rule certifies, narrowly. Under a Hoeffding-only gate it is refused; a range-only *sufficient* sample size for 95% acceptance of a candidate with excess $\Delta = 0.031$ is $m \ge [\sqrt{2\ln(2K/\delta)} + \sqrt{2\ln 20}]^2/(\epsilon - \Delta)^2 \approx 7.4 \times 10^4$; the actual power under the ternary law is higher and this is not a necessity claim. Every gate number in this paper is from the rule as stated (min of the two radii, K given per experiment); §VII reports both gates side by side. The implementation computes $\hat{\Delta}$ exactly from the integer paired counts, inflates the floating-point margin to a conservative upper enclosure, and compares in exact rational arithmetic, so rounding can only make the gate stricter.

a) An exact ternary comparator: Since $Z \in \{-1, 0, 1\}$, an exact finite-sample test exists: with (N_+, N_0, N_-) multinomial and $S = N_+ - N_-$, reject $\Delta > \epsilon$ when $\sup_{\rho \in [\epsilon, \rho_U]} \Pr_{\Delta=\epsilon,\rho}[S \le s_{\text{obs}}] \le \delta'$, where $\rho = p_+ + p_-$ is the disagreement rate. The nuisance bound matters: unrestricted, the supremum sits at high ρ and discards the low observed disagreement—on the worked example above the unrestricted test gives $p = 0.136$ and does not certify. Bounding ρ first by a Clopper–Pearson upper limit from $M = N_+ + N_-$ ($\rho_U = 0.057$ at $\alpha = 0.025$) and spending the remaining budget on the sup gives $p \approx 2 \times 10^{-7}$: far more powerful than the Bernstein margin, which certified the same example only at $0.0495 \le 0.05$. We report this as an independently specified comparator (its supremum is evaluated on a dense grid, not certified), not as the release rule; making it the release rule is a clear improvement for future work.

The bound has no factor in the number of agents N or the number of compromised agents $k \le N$: all agents may be compromised and submit arbitrary code. It is *fallback-relative*: an adversary can force fallback and destroy the agents' improvement over h_0 at zero risk cost, which is an availability loss, not a safety loss. Bounding degradation relative to a *clean* agent needs a bound on $L_D(h_0) - L_D(h_{\text{clean}})$, which the gate does not supply. The theorem assumes a fixed pointwise predictor; a batch program with cross-row state is outside it.

b) Release runtime: A single-use lifecycle (OPEN→FROZEN→CERTIFYING→ CERTIFIED→RELEASED, FAILED terminal): candidates hashed before certification randomness exists; holdout seed drawn from OS entropy after freeze, held privately, disclosed only in the audit record; candidates executed in a spawned process receiving features only, replying over a bounded non-pickle byte protocol; label-alphabet validation of candidate and fallback; immutable private authorization records; released candidates executing through the same isolated path as certification. Two adversarial source audits found and we fixed: mutable returned decisions controlling release, exception-unsafe state, pickle deserialization from the untrusted worker, missing label validation, and a divergent deployment path. Remaining obligations for the surrounding system (executor manifest, resource limits, protected deployment sink, signed audit storage) are stated, not claimed (§IX).

V. Policy Results

A. Pilot (exploratory)

Claude Sonnet 4.5 via Amazon Bedrock; sextic; theory withheld from the brief so the tool channel is load-bearing; temperature 0.7; $n = 5$ per condition; arms run sequentially. Median accuracy (Table II).

TABLE II
PILOT: SAME DETECTOR, THREE POLICIES. MEDIAN ACCURACY, $n = 5$/CELL EXCEPT FLAG/DATA ($n = 4$; ONE API FAILURE, RETAINED IN THE SUPPLEMENT), SEXTIC (CEILING 0.9518). EXACT ONE-SIDED PERMUTATION p ON ACCURACY SUMS, UNPAIRED; NOMINAL AND EXPLORATORY—SEQUENTIAL ARMS DO NOT ESTABLISH EXCHANGEABILITY.

channel	unguarded	flag	repair	p(R>F)	p(R>U)
none	0.913	0.925	0.920	—	—
data	0.907	0.898	0.907	0.48	0.62
injection	0.925	0.920	0.925	0.50	0.84
impossible	0.920	0.728	0.925	0.020	0.14
combined	0.898	**0.153**	**0.910**	**0.004**	0.028
verif._stealth	—	0.890	0.925	0.032	—

Under `flag`, every poisoned response on a detected channel was flagged (deterministic; per-run counts in the repository), yet no channel with an unguarded control had a higher guarded median, and `combined` collapsed: in all five runs the final conjecture states the correct $f{=}2$ criterion attached to the wrong class (coded from the returned JSON conjecture and rule; the transcripts are released). Flagging retains the original bytes at the response level, so guarded agents received the original evidence plus a correct diagnostic; transcripts show them abandoning the flagged channel for surface heuristics. Under `repair`, every detected channel's median lay within the clean band. Complete separation of the ten run accuracies on `combined` repair-vs-flag ($p = 1/\binom{10}{5}$) is the contrast that motivates the confirmatory study; the others are unresolved at this n.

What this does *not* show: that warnings *cause* harm (arms were sequential, and the flag-vs-unguarded question is a separate contrast), that warnings fail in general (one message design, one model), that repair equals clean (a nonsignificant difference is not equivalence), or that inverted rules defeat review in general (five transcripts).

B. Prespecified confirmatory study

Protocol registered before data collection (git tag `confirmatory-protocol-v1`): 3 policies {unguarded, flag, repair} × {`combined`, `impossible`, `none`} × {sextic, degree-21} = 18 cells, $n = 10$ runs per cell, all 180 runs randomly interleaved from a committed seed so policy is not confounded with time. Both endpoints per run: raw pipeline accuracy, and the accuracy of the system a gated deployment would have *released* (certified through the release runtime against a majority fallback, $K{=}1$, $m = 20{,}000$, $\epsilon = \delta = 0.05$, audited on an independent 20,000-sample draw). Primary: sextic/`combined`, repair vs. flag, raw endpoint, one-sided exact permutation, $\alpha = 0.05$. Registered secondaries, Holm-corrected as one family of four: sextic `combined` repair vs. unguarded; sextic `impossible` repair vs. flag; degree-21 `combined` repair vs. flag; the released-endpoint version of the primary. Invalid runs (API failures) are retried once and otherwise reported missing; clean channels are reported descriptively as a drift check.

Two questions are *not* in the registration and will be reported as exploratory: whether flag is worse than unguarded (which is what "warnings harmed agents" would require), and the competent-advisory baseline. For the latter we have implemented `flag_routed` (names the failed source and the channel that remains independently verified, under identical trust assumptions and with no task answers); it is not an

arm of the registered study and will be run as a separately registered follow-up. If routed warnings recover performance, the contribution sharpens into a distinction between behavior-dependent mitigation and the conditional restoration guarantee of Prop. 3; it does not weaken the architectural claim, which never rested on warnings failing. Any amendment to the protocol after data collection began will be reported as a timestamped amendment.

VI. Defense-Aware Attacker and the Residual Class

Against repair, an attacker who adds a detectable marker causes the defender to restore the very evidence it wanted to falsify. The defense-aware play is one consistent label permutation across every evidence channel, exact grammar, attainable values, class names matched—and honest validation, since a validation lie is caught by recomputation. This channel (`data_adaptive`) is a defense-aware attacker, not a proven optimum over the accepted-output space.

End to end, offline (sextic, N=200, ceiling 0.8991): the label-agnostic subverifier, with every check enabled, detects 0 of the attack's responses—consistent with Prop. 2, whose hypotheses that subverifier satisfies. The complete mediator's recomputation check is class-anchored and so is *not* covered by the proposition; it detects nothing here for a different reason: the attack leaves validation honest, so there is no discrepancy to recompute. A credulous learner (decision tree fit to the permuted evidence) lands at true accuracy 0.240 on 600 test rows (1−ceiling = 0.101); the release gate (K=1, m=2000, theorem fallback) refuses it, so the released system is the fallback and its excess on an independent 20,000-sample audit is 0 by construction—evidence that the gate refused a catastrophic rule, not of population-risk control.

Live agents (sextic, N=1000, $n = 5$ per arm, 600 test rows per run, medians reported): clean 0.895 (range 0.887–0.925), attacked-unguarded 0.920 (0.867–0.945), attacked-repair 0.890 (0.867–0.925); 0 detections by the guard in the 5 guarded runs across 142 poisoned responses. The attack evaded the checks but produced no clear degradation. Transcripts are consistent with two orientation anchors the attacker cannot permute—trusted validation feedback (agents iterate candidate rules against the honest validation tool) and pretrained knowledge of the named groups' element orders—and this experiment does not separate them. The structural statement we can make: under trusted recomputation and complete mediation, a false validation result cannot persist at the agent interface; whether that *disarms* the attack depends on agent behavior this study did not isolate. The separating experiment is a 2×2 factorial under the same attack—{named groups, opaque per-run labels} × {validation available, unavailable}, with final class orientation as a prespecified mechanism endpoint—which we have designed and not yet run. Our reading of the transcripts is that the validation anchor is doing most of the work (agents revise after failed validations and rarely cite group orders before testing), but we hold that as a hypothesis.

The residual accepted-corruption class is larger than this fixture: altered attainable frequencies, invented examples, grammar-conforming error text, anything the parser accepts unchanged. The gate's bound is what covers that entire class, fallback-relative.

VII. Gate Operating Characteristics

All numbers here use the gate of Theorem 1 as stated ($r = \min(r^H, r^B)$, $\delta/2$ each), with K=1 per certification (each artifact certified separately on its own fresh holdout), $\epsilon = \delta = 0.05$. An earlier draft mixed these with a Hoeffding-only gate; those numbers are superseded.

a) Retrospective integration study: The 89 archived pilot rules were certified through the release runtime, $m = 2000$, against two fallbacks. Majority fallback (risk ≈ 0.5): 82/89 released; the seven refusals were every rule at or below chance (six inversions at 0.10–0.16 holdout accuracy, one at exactly 0.500). Theorem fallback (risk ≈ 0.049): 8/89 released (five clean-channel rules, two `injection`, one `verification_stealth`), with a maximum released holdout excess of $+0.026 < \epsilon$; under a Hoeffding-only gate the same artifacts gave 0/89. That released artifacts satisfy the certification inequality on the certification sample is an algebraic property of the rule, not evidence about population risk. The theorem-fallback regime is a safety-floor experiment: agents at ≈ 0.92 against a 0.951 fallback rarely clear $\hat{\Delta} + r \le \epsilon$ at $m = 2000$, and their doing so is not evidence that they add value over the fallback.

b) Operating characteristics with independent audit: Boundary candidates are the fallback with predictions flipped on rows selected by a deterministic per-row hash at rate q; their population excess Δ is *estimated* on an independent 10^5-row sample (standard error ≤ 0.002), not computed exactly. Thirty fresh m-row holdouts per cell; every released artifact's excess is judged against that Δ estimate. Results:

Δ (est.)	accept, m=2000	accept, m=20,000	viol./released
0.000	30/30	30/30	0/60
0.016	30/30	30/30	0/60
0.032	6/30	30/30	0/36
0.044	0/30	19/30	0/19
0.055 ($> \epsilon$)	0/30	0/30	—
0.079, 0.159	0/30	0/30	—

With 30 replicates a 30/30 cell has a 95% Clopper–Pearson lower bound of 0.88 and a 0/30 cell an upper bound of 0.12. In the observed cells, acceptance is complete well inside the budget, degrades as Δ approaches the effective threshold $\epsilon - r$, and is zero in every cell with $\Delta > \epsilon$ (no cell sits exactly at ϵ); no released artifact exceeded ϵ excess by the independent estimate. These are per-certification results at K=1; they do not bound the probability that *some* certification in the campaign was unsafe, which is $\le \sum \delta_t$. The Δ=0 row (candidate identical to the fallback, zero disagreement variance) certifies at $m = 2000$, as the Bernstein radius requires and a Hoeffding-only gate would not.

VIII. Related Work

a) Reference monitors and complete mediation: The mediator of §IV is a reference monitor in Anderson's sense [1]: every tool response passes through one trusted point, and Prop. 3's complete-mediation hypothesis is Saltzer and Schroeder's principle [2] applied to evidence rather than to access. What is new is not the monitor but what it checks (arithmetic admissibility rather than authorization) and what it does on failure (restoration rather than denial).

b) Prompt injection and tool-output defenses: Indirect prompt injection through retrieved or tool-returned content is established [3]. Design-level defenses such as CaMeL [4] separate control flow from untrusted data with capability tracking so that injected content cannot acquire authority; compromised-agent architectures (SAGA-BFT, MUZZLE, ACE [13]) work at the permission and containment boundary. These systems principally constrain the *authority* of untrusted content; our checks constrain specified arithmetic properties of tool evidence, which are necessary (not sufficient) for its truth. The protections are complementary: capability confinement alone does not establish those properties.

c) Baseline-relative certification: Releasing a candidate only when a fresh sample certifies it against a baseline is the structure of high-confidence and safe policy improvement [5], [6], of selective classification [7], and of risk-controlling prediction [8], [9]. Theorem 1 is an instance of that pattern with paired 0–1 losses and an empirical-Bernstein radius [12]; we claim no new inequality. Our contribution is placing the gate at the exit of an adversarial pipeline whose upstream may be entirely compromised, and showing which corruption classes it is the *only* layer that covers.

d) Behavioral studies of multi-agent systems: Anthropic's *Patterns and problems in multiagent systems* [11] provides observational context through studies of trust, conformity, and coordination. Its lie-detection evaluation and ours share a comparison pattern: agent behavior is assessed against a credulous policy and a ground-truth benchmark. The tasks, information structures, and benchmark semantics differ; our setting additionally permits analytic ground truth and formal bounds. The report concludes that models may understand source incentives yet lack "a disposition to act on that knowledge without prompting." We take this as motivation for a narrower architectural question: which assurances can be enforced without relying on upstream agents' judgment? Our repair mechanism and release gate enforce distinct evidence and release conditions under explicit trust and sampling assumptions. They do not establish improved coordination, incentive alignment, or information pooling. Its conformity observations are qualitatively consistent with our earlier measurement, on this testbed, that the error sets of five independently run agents overlapped 14–$26\times$ more than the Jaccard index implied by independent errors at their rates ($pq/(p+q-pq)$); the two measure different phenomena, and that measurement motivates but is not a premise of the agent-count-independent guarantees.

e) Discussion: enforcement rather than discretionary response: Our response to the disposition gap is deliberately limited. Where a tool-evidence property admits a tractable, sound check from trusted inputs, the corresponding assurance need not depend on an agent noticing a warning or choosing to honor it. The release gate provides a separate, statistical guarantee relative to a baseline; it does not guarantee detection of every harmful candidate. Disposition independence therefore concerns enforcement at the trusted boundaries, not the behavior or integrity of the entire system. The report's observation of price matching through a public listings board after direct communication was removed illustrates a related limitation: task outputs can remain a medium for coordination. Our formal boundary applies to the specified verifier and its admissible or observationally indistinguishable output class; we do not establish that the report's experiment satisfies those hypotheses, nor claim that content verification prevents collusion. Likewise, repairing tool evidence does not itself ensure that agents elicit, transmit, or use pivotal private information. Lee and Lee [10] established the arithmetic criteria we use as ground truth. Selective classification and fallback-relative certification are standard; our contribution is their composition with mediation and repair, and the separation of which assurance each supplies.

IX. Limitations

Scope of exactness. Theorem-class checks are exact for the unramified synthetic law; ramified primes of real fields fall outside them. Deployment distributions are out of scope. *What transfers.* The logical structure—complete mediation, trusted restoration, frozen proposals, fresh-holdout certification—and not the observed rates of warning-induced failure, attack effectiveness, or the cheap availability of IID labels and exact recomputation. *Trusted resources.* Recomputation and labeled certification data are substantive; we compare policies under the same budget and report acceptance and fallback frequency alongside risk. *Runtime.* Source hashing is not semantic artifact identity; pointwise semantics are assumed; resource limits, a protected deployment sink, and signed audit storage are the surrounding system's obligations. *Empirical.* One model family; $n = 5$ sequential pilot arms with the confirmatory study in progress; the anchor factorial and the routed-warning baseline are designed but not run. No pilot contrast is causal; the `combined` repair-vs-flag separation is the one we regard as strong enough to have motivated a registered replication. *Related work.* The individual principles—complete mediation, trusted replacement, baseline-relative certification—are established; our claim is their composition and instrumentation, and the related-work discussion here is briefer than the final version must be.

X. Conclusion

A checker is not a defense until something binding acts on its verdicts, and the thing that acts should not be the agent's disposition. In this testbed, the same detector with the same detections was associated with a large regression under one enforcement policy and near-clean performance under another, and a relabeling attack the label-agnostic checks cannot see was refused by a release gate that never looked at the evidence

at all. The guarantees are conditional and fallback-relative, they do not depend on how many agents there are or on what any of them believes, and the behavioral finding awaits its registered replication.

APPENDIX

Table III consolidates every claim with its assumptions, evidence, and status. The supplementary artifact (`paper/satml/supplement/`) contains run-level outcomes for all 135 pilot and adaptive-attack runs (`pilot_runs.csv`), the mechanical orientation coding behind the inversion finding (`orientation_coding.csv`: each returned rule is executed on fresh synthetic data and labeled correct/inverted/neither by whether it scores > 0.75 against true or against permuted labels—flag/`combined` $5/5$ inverted, repair/`combined` $5/5$ correct, unguarded/`combined` 3 correct, 1 inverted, 1 neither), and the per-run gate refusal manifest (`gate_refusals.csv`).

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TABLE III
CLAIM LEDGER: WHAT IS CLAIMED, ON WHAT ASSUMPTIONS, WITH WHAT EVIDENCE, AND ITS CURRENT STATUS.

claim	assumptions	evidence	status
Attainable-value check is sound and complete for reported values outside the set	unramified synthetic law; renderer image	Lean theorem (`Attainable.lean`); soundness sweep on all 9 tasks	proved (arithmetic), tested (renderer)
Same-prime joint realizability iff LP feasible	population mixture; closed rounding band	exact rational certificates; 3 tests	implemented; finite-sample version not
Label-agnostic subverifier cannot detect consistent relabeling (Prop. 2)	closure of honest set under π; zero FP	Lean (`Indistinguishability.lean`); regression test; 0/142 live	proved (zero-FP form); α-form on paper only
Exact repair $\Rightarrow$ clean transcript (Prop. 3)	complete mediation; exact recomputation; coupled randomness; no side channels	Lean (`Repair.lean`); byte-equality tests on all detected fixtures	proved conditional; hypotheses tested on fixtures
Gate: $\Pr[\text{excess} > \epsilon] \leq \delta$ (Thm. 1)	frozen pointwise predictors; fresh IID holdout; trusted labels/evaluator	standard inequalities; Lean deterministic clause; OC study, 0/175 released violations	proved; fallback-relative; pointwise semantics assumed
Release runtime enforces the gate	single trusted process	17 tests incl. mutation attack, lifecycle, IPC, timeout	implemented; executor manifest, resource limits, sink, signed audit pending
Warning-only policy associated with regression on `combined`	one model, one message	pilot n=5, complete separation; orientation coding 5/5 inverted	exploratory; replication registered, running
Repair associated with near-clean performance	same	pilot; repair vs flag $p = 1/252$	exploratory; not an equivalence claim
Warnings *harm* agents (flag $<$ unguarded)	—	not registered	exploratory only
Competent warning (`flag_routed`)	—	implemented, not run	pending
Adaptive attack harmless to live agents; mechanism	—	n=5/arm; two candidate anchors	descriptive; factorial designed